Comparing LLM and Human Behavior in Argumentation-based Reasoning Puzzles

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Abstract

We propose a user study to evaluate the ability of human participants to solve argumentation-based reasoning puzzles and compare their performance with that of Large Language Models (LLMs). The study consists of two parts. In the first part, reasoning puzzles concerning witness testimony, generated using an automated library originally developed to evaluate LLM reasoning, are presented to both humans and LLMs. We systematically vary properties of the underlying argument graphs, including their size, their shape, the order in which arguments are presented, and the inclusion of irrelevant argument chains, to investigate how these factors affect the reasoning performance of humans and LLMs. In the second study, we evaluate human and LLM performance on puzzles derived from manually constructed argument graphs with more realistic and natural attack relations, albeit at the cost of increased linguistic complexity. A particular focus of both studies is the evaluation of cognitive biases in both humans and LLMs.

Keywords

Argumentation, Reasoning tasks, Large Language Models, Cognitive biases, User study

1. Introduction

Large Language Models (LLMs) have demonstrated the ability to solve reasoning tasks for which they were not explicitly trained [1]. A variety of benchmarks have been developed to evaluate these reasoning capabilities [2], and LLMs have been shown to exhibit certain cognitive biases [3]. However, the extent to which their reasoning behavior and biases resemble those of humans remains under-explored.

This paper therefore proposes an experimental design to compare human and LLM reasoning and performance on argumentative puzzles. The proposal consists of two studies, one with puzzles based on an existing benchmark and one with more complex, but more realistic puzzles. We also introduce a pilot study that will be conducted before Study 1 at the CERCA workshop. Before describing the experimental design and procedure, we first provide the necessary formal preliminaries, context on the existing reasoning puzzles, and related research on cognitive biases in LLMs and cognitive research in argumentation.

2. Background

In this section, we provide background on the formal preliminaries, the argumentation-based reasoning benchmark used in the first part of our study, related work on cognitive argumentation experiments, and cognitive biases in LLMs.

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2.1. Formal Preliminaries

An *abstract argumentation framework* (AF) [4] is a pair $\mathcal{AF} = (\mathcal{A}, \mathcal{R})$, where $\mathcal{A}$ is a set of arguments, and $\mathcal{R}$ is a binary attack relation on these arguments. For every $(A, B) \in \mathcal{R}$, A attacks B. AFs can be represented as a directed graph, where the nodes represent arguments and the arrows represent the attack relations (Figure 2). An argument *reinstates* another argument if it attacks an attacker of that argument; B defends A if B attacks C and C attacks A. Given an AF, sets of arguments that are collectively *acceptable* can be determined [4].

Definition 2.1. Let $\mathcal{AF} = (\mathcal{A}, \mathcal{R})$ and, $A \in \mathcal{A}$, and a set of arguments $S \subseteq \mathcal{A}$. Then:

- A is *acceptable* with respect to S iff each argument attacking A is attacked by any $B \in S$;
- S is *conflict-free* iff there exists no $(A, B) \in \mathcal{R}$ with $A, B \in S$; and
- S is *admissible* iff its is conflict-free and every $A \in \mathcal{A}$ is acceptable w.r.t S.

2.2. Argumentation-based Reasoning Puzzles

Previous work set out to create a benchmark for evaluating the reasoning capabilities of LLMs that mitigates data contamination (test data being included in the training data), scales to future models, and provides formally unambiguous interpretations [5]. The result is a library called *Reasoning-Benchmarks* [1], which dynamically generates reasoning puzzles of varying complexity based on formal argumentation in natural language. The puzzles consist of an instruction, a story, and a question that must be answered based on both the instruction and the story. We use these puzzles as the basis for our reasoning tasks.

The *Reasoning-Benchmarks* library generates an abstract argumentation framework and translates it into a natural language prompt (Figure 1). Specifically, the AF is constructed from a set of predefined parameters, including the number of arguments and the graph shape. The AF is then translated into a natural-language reasoning puzzle via a prompt, using an ontology consisting of statements and names combined with additional prompt parameters.

For the AFs, two graph topologies are supported: chain graphs, in which each argument attacks the preceding argument, and star graphs, in which multiple argument chains attack a common central argument. Figure 2 shows an example chain graph and a corresponding prompt, and Figure 3 shows an example star graph and an associated prompt.

The puzzles concern witness testimony, where each argument represents a witness and their statement. In each statement, a witness undercuts the believability of another witness. Two tasks are supported: a *binary task*, which asks whether the main witness statement can be believed (e.g., Figure 2b), and a *multi-label task*, which asks which witness statements can be believed (e.g., Figure 3b). Solving either task requires identifying the accepted arguments in the underlying argument graph (Definition 2.1).

In the reasoning puzzles, names and main statements in the prompts can be generated dynamically, as can the attack word that defines the attack relation between arguments (e.g., "lying" in Figure 2b and

[1] https://pypi.org/project/reasoning-benchmarks/

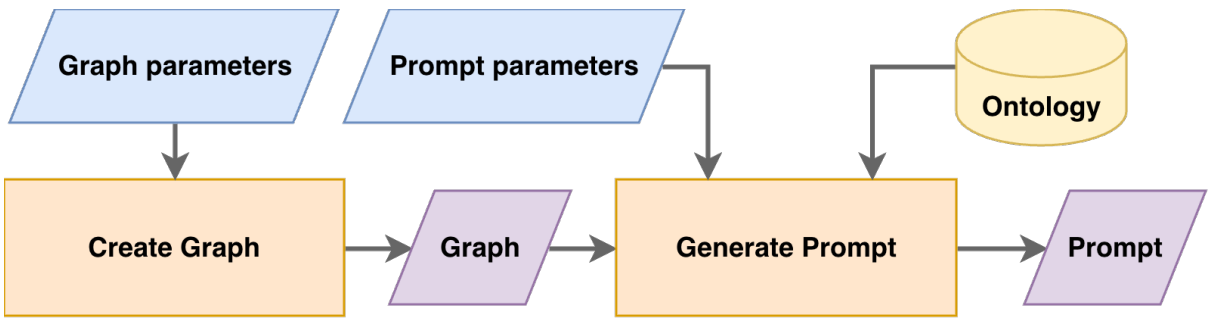
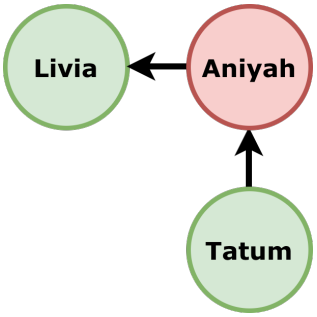


Figure 1: Reasoning Benchmark pipeline to generate reasoning tasks.

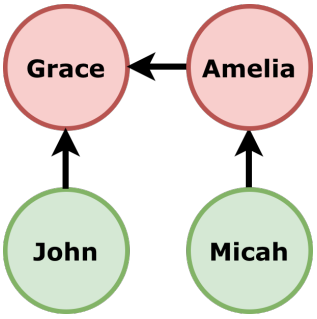


(a) Chain graph

The following is a reasoning puzzle. Witnesses should always be believed unless there is believable testimony that they are lying. Now consider the following facts:
Witness Livia says that the employee arrived on time.
Witness Aniyah says that Livia is lying.
Witness Tatum says that Aniyah is lying.
Final question: can we believe that the employee arrived on time?
End your answer with: "Answer: yes or no".

(b) Binary prompt

Figure 2: Example of an argument chain with three arguments (a) and its associated binary prompt (b). Accepted arguments are shown in green and rejected arguments in red.



(a) Star graph

The following is a reasoning puzzle. Witnesses should always be believed unless there is believable testimony that they are not credible. Now consider the following facts:
Witness John says that Grace is not credible.
Witness Micah says that Amelia is not credible.
Witness Grace says that the password was changed.
Witness Amelia says that Grace is not credible.
Final question: which witnesses can be believed?
End your answer with:"Answer: list of names that can be believed".
E.g. Answer: ['Alice', 'Bob']

(b) Multi-label prompt

Figure 3: Example of an argument star with four arguments (a) and its associated multi-label prompt (b). Accepted arguments are shown in green and rejected arguments in red.

"is credible" in Figure 3b). In addition, arguments may be presented either in the order based on the graph, as in Figure 2b, or in a shuffled order, as in Figure 3b. Finally, prompts may include irrelevant arguments or chains of arguments that are not connected to the main argument structure. For example, adding the statement "Phoenix says that Miles is lying" to the binary prompt in Figure 2b does not affect the outcome. For the multi-label task, such irrelevant arguments should be evaluated. Both shuffling and the inclusion of irrelevant arguments may increase task difficulty.

2.3. Cognitive argumentation experiments

The current study is based on findings and methods from existing experiments cognitively investigating computational argumentation. The first major study in this field was performed by Rahwan et al. [6], who tested how humans evaluate reinstatement (see Section 2.1). In computational argumentation, a reinstated argument is equally acceptable as an unattacked argument. In their study, Rahwan et al. [6] found that participants use reinstatement when evaluating arguments. But the confidence in the conclusion of the original argument is reduced compared to when it was not attacked. This is relevant for the current study, because reinstatement is used in the evaluation of most puzzles. For example, in Figure 2, Tatum reinstates Livia's original claim. However, in [6] reinstatement was only evaluated for chains of 3 arguments, and since reinstatement reduced confidence in the original argument somewhat, it is possible that this effect will compound for longer chains.

Following [6], several empirical cognitive studies on computational argumentation have been conducted [7]. Of those studies, most have focused on how humans evaluate and interpret arguments and attacks between them. In [8], the cognitive plausibility of various argumentation semantics was tested using two experiments. Argumentation semantics specify how to determine which sets of arguments

can be considered acceptable, based on their attack relations. In the first experiment, participants were, in groups, tasked with drawing all attack relations between arguments in a set. With this experiment, they validated that participants correctly identified attack relations between arguments in a given argumentation framework. They also found that conflicting arguments can be created that people interpret as unidirectional attacks, especially undercutting the trustworthiness of a source is convincing as a unidirectional attack, which is very similar to the attacks used in this study. In the second experiment, participants individually judged the acceptability of arguments and then discussed their answers in a group before making a final judgement. The researchers compared the acceptability of arguments as rated by participants to their acceptability under various argumentation semantics [4], and that semantics that minimize being undecided on the acceptance of arguments were most comparable to participant judgments. Meaning that participants minimized uncertainty while evaluating arguments. When comparing human behaviour on argumentation tasks to LLM behaviour, these findings are relevant to consider, since LLMs might have a different need for certainty than humans, and consider the directionality of attack in another way than humans, leading to a different interpretation of AFs.

In Scheffers et al. [9], participants were shown stories based on argumentation frameworks and asked to explain one argument using the other arguments in an AF. Explanations could contain any number of arguments. The most frequently used explanation consisted of the direct defender of the explained argument, and participants rarely included unrelated arguments in their explanations. Showing that they could determine the most relevant information and use this in explanations, even when the order of arguments was shuffled randomly.

From these prior studies, we take inspiration for the phrasing of unidirectional attacks [8], varying contexts to reduce context effects [10], and the random shuffling of arguments [9]

2.4. Cognitive Biases in LLMs and Humans

Systematic deviation of decisions from a normative standard or a task's stated objective, namely cognitive biases, is common among humans. Such biases arise from cognitive limitations on rational deliberation [11], and some heuristics are understood to be adaptive under conditions of bounded rationality [12]. No such justification extends to AI systems, which are designed to produce accurate, rational responses in alignment with user objectives. Yet, trained on human-generated text and optimised to produce human-like outputs, LLMs inherit systematic inferential biases [13], acquired from regularities in training corpora and reinforced through human feedback during training or fine-tuning [14, 15].

Empirical research has documented a range of cognitive biases in LLM outputs, mirroring those observed in human judgment and decision-making. These include anchoring effects—the tendency to insufficiently adjust estimates away from an initial reference value [16, 17], framing effects—predictable shifts in responses driven by how a problem is presented [17, 13], and priming effects—where prior exposure to certain stimuli unconsciously shapes subsequent decisions [15], among others.

One type of cognitive bias that is closely related to the task objective of our proposed studies is order effect. Qiao et al. [18] find that GPT exhibits a recency effect in legal judgment tasks—where judgements are more strongly influenced by later presented evidence items than earlier ones—across all experimental conditions, including those in which the effect is eliminated in human participants, demonstrating that LLMs do not merely replicate human cognitive biases, but amplify them.

The proposed study extends this line of inquiry to argumentation. We will not only compare human and LLM performances for solving argumentation puzzles, but also investigate whether LLMs display order effects when processing argumentative tasks by manipulating argument presentation order, and how such biases are manifested in a similar or different way to human participants.

3. Methods

We propose to evaluate the ability of human participants to solve argumentation-based reasoning tasks and compare their performance to that of LLMs using two studies.

Table 1
Parameter settings of the data subsets used in the experiment.

Setting	Subset A	Subset B	Subset C
Chain range	2–10	2–10	2–10
Star range	3–10	3–10	3–10
Shuffled	No	Yes	Yes
# Irrelevant chains	0	0	1–3
Length irrelevant	0	0	1–3
Total cases	96	96	96

In Study 1, we will use reasoning puzzles generated by the Reasoning-Benchmarks library to assess human performance. These tasks were originally designed to evaluate LLM reasoning and can be scaled indefinitely in complexity. We will compare human performance with LLM performance (The results of LLM performance can be found in Section 4).

A limitation of the Reasoning-Benchmarks library is that attack relations are instantiated using a single, fixed relation type (e.g., "lying"), which can be repetitive and less realistic. Therefore, in Study 2, we will examine both human and LLM performance on a smaller set of more realistic argument graphs, in which attack relations are more representative of real texts and decisions participants encounter in daily life.

3.1. Study 1

3.1.1. Materials

We will use the Reasoning-Benchmarks library to generate 96 puzzles comprising all unique argument chains of lengths 2 to 10 and all unique star graphs containing 3 to 10 arguments. Based on these 96 puzzles, we construct three experimental subsets. In Subset A, arguments are presented in their original logical order. In Subset B, arguments are randomly shuffled while ensuring that their ordering differs from that of Subset A. Subset C further increases task complexity by combining argument shuffling with the inclusion of irrelevant arguments. Specifically, each puzzle is augmented with one to three randomly generated irrelevant argument chains, each consisting of one to three arguments. This results in a total of 288 puzzle variations, as shown in Table 1. We will evaluate both the binary and multi-label tasks described in the background section across all puzzle subsets.

3.1.2. Design

To compare human and LLM performance on argumentation-based puzzles, we use two reasoning tasks (labelling and binary choice, mirroring the multi-label and binary tasks for LLMs respectively) and three subsets (see Section 3.1.1). These subsets cover five main variables: 1) the typology of the graphs (chain or star), 2) the number of arguments in the chain/star (range), 3) the presentation order of arguments (in graph order or shuffled), 4) the inclusions of irrelevant arguments, and 5) the number of irrelevant arguments included.

3.1.3. Pilot

A pilot study will be conducted to establish the time participants need to complete the reasoning puzzles and to determine the optimal presentation format. The pilot will inform both the number of puzzles completable within a 10-minute window and, combined with a power calculation, the required sample size.

During the pilot at CERCA, we plan to give every participant 10-15 minutes (depending on the workshop schedule). In this time, participants will read the information sheet and instructions and complete ten reasoning puzzles (the total number of puzzles might be adjusted based on available time in the workshop). The puzzles will be of different lengths to assess a realistic maximum number of

arguments in reasoning puzzles. Time taken for each part will be recorded. After participants complete the task, we would like to ask them some questions about their experiences with the experiment and their approach to solving the puzzles. This will allow us to improve the experiment and to better understand how people approach solving the reasoning puzzles.

3.1.4. Procedure

Puzzles will be presented to both humans and LLMs. Humans will be recruited via Prolific (https://www.prolific.com). Participants will first provide informed consent, then receive instructions, after which they will start the experiment. Participants will receive a number of puzzles that will be determined based on the pilot. This will be the average that participants in the pilot complete in 10 minutes. Puzzles will be presented in two blocks, one with the binary task, one with the multi-label tasks. Both tasks and blocks will be presented in random orders.

LLMs were presented with prompts, such as the ones shown in Figures 2 and 3. We collected both their final answer as well as their full output, some of which contained reasoning or justification.

3.1.5. Analysis

The paired datasets produced by LLMs and human participants solving the same set of reasoning tasks enable a direct comparison of their behavior. This allows us to quantify differences in overall task performance and to assess the extent to which LLMs approximate human reasoning under identical conditions.

The first analysis will focus on aggregate performance: we will compare how well humans and LLMs solve the tasks by looking at the percentage of correct responses, both overall and at the individual level and for the binary and multilabel tasks. This analysis will include identifying which types of questions are systematically difficult for humans versus those that are challenging for LLMs by looking at accuracies.

We will also explore how the variations between the puzzles affect performance. In particular, we examine the impact of graph-related dimensions such as the number of arguments, graph type (chain or star), graph shape, and the overall complexity of the graphs. This enables us to detect how sensitivity to combinatorial and structural factors affects human decision-making and how it differs from LLMs.

In addition, we will evaluate how variations in the prompt structure—specifically, order shuffling of arguments (subset B) and the inclusion of irrelevant argument chains (subset C)—influence human and LLM performance. We will conduct separate 2 (Human vs. LLM; between-subject) × 2 (Subset A vs. Subset B vs. Subset C; within-subject) mixed ANOVAs. This allows us to probe potential order effects, including primacy and recency effects, and the ability to discard irrelevant information in both humans and LLMs.

Finally, we will conduct a one-way ANOVA to compare multiple LLMs with varying reasoning effort levels to human performance. A non-significant difference would indicate that a model and reasoning configuration closely aligns with human behavior.

3.2. Study 2

The puzzles generated by Reasoning-Benchmark are created to be solvable by LLMs, and thus comparing human and LLM performance and reasoning on these benchmarks might favor LLMs. In Study 2, we will compose more diverse puzzles that are closer to natural language. While more realistic, they could potentially be more difficult due to the diverse content of the arguments and attacks. These stories will be similar to those used in [6], [8], and [9] (see Example 3.1).

Example 3.1. An example from [19] of a more natural scenario for the AF introduced in Figure 2.

A) Jessie saw Steven at the crime scene yesterday at the time of the crime. Therefore, Steven is guilty.

Table 2
Mean LLM accuracy per subset for the binary and multi-label tasks.

Task	Subset A	Subset B	Subset C
Binary	89.1%	87.9%	79.0%
Multi-label	83.3%	81.8%	66.7%

B) Steven says that he has never been to the crime scene. Therefore, Steven couldn't have been seen at the crime scene at the time of the crime.

C) There is a picture of Steven at the crime scene two years ago, so Steven is lying when he says he's never been to the crime scene.

The analysis of Study 2 will focus on more realistic argument content, allowing us to assess whether increased realism improves performance by making the tasks more natural or degrades performance due to increased complexity.

4. Initial LLM results

We have conducted part of Study 1, where we investigate the performance of LLMs on the three subsets described in Table 1. Specifically, we tested with GPT-5.4, GPT-5.4-mini, and GPT-5.4-nano, which are, at the time of writing, the state-of-the-art models from OpenAI [2]. Each of these three models supports "reasoning effort" parameter, which refers to the amount of compute allocated to internal reasoning during task execution [3]. We varied this parameter across the five possible values: none, low, medium, high, and xhigh.

Figure 4 shows the mean accuracy of the three LLM models across the five reasoning effort values on both the binary and the multi-label tasks. Performance generally increased with a higher reasoning effort. Interestingly, the best performance on the binary task was achieved with the smallest model (nano) on the highest reasoning effort setting (Xhigh).

Table 2 displays the aggregated performance of the three models and all reasoning efforts across the three different subsets as described in Table 1. While the lower accuracy on subset B compared to subset A was minimal, a substantial decrease in performance in subset C was observed. This suggests that the shuffling of the arguments did not have a large impact on the performance of LLM models, but the combination of order shuffling and the inclusion of irrelevant arguments did.

5. Discussion

This paper proposes two studies to compare human and LLM performance and biases on reasoning puzzles, by varying AF characteristics, argument order, and the inclusion of irrelevant arguments. Study 1 will employ an existing benchmark designed for LLMs, and Study 2 will employ reasoning puzzles designed for humans. Initial LLM results in Table 2 suggest that the effect of shuffling the arguments alone on LLM performance is minimal, but the inclusion of irrelevant arguments decreases accuracy. Based on previous work on humans [9], we predict that neither the order of the arguments nor the inclusion of unrelated arguments would affect their performance.

The design of the studies has several potential limitations, which we aim to mitigate by looking at the results from the pilot. First, this will be a 0-shot task for LLMs, meaning a single prompt is given to an LLM and an answer is returned. In an experiment with humans, a similar setup where participants only complete a single trial is impractical, because a large number of participants would be required, and every participant would have to spend time reading instructions and providing consent. When giving participants multiple trials, other factors—such as learning and fatigue effects—might create

[2] https://platform.openai.com/docs/models
[3] https://platform.openai.com/docs/guides/reasoning

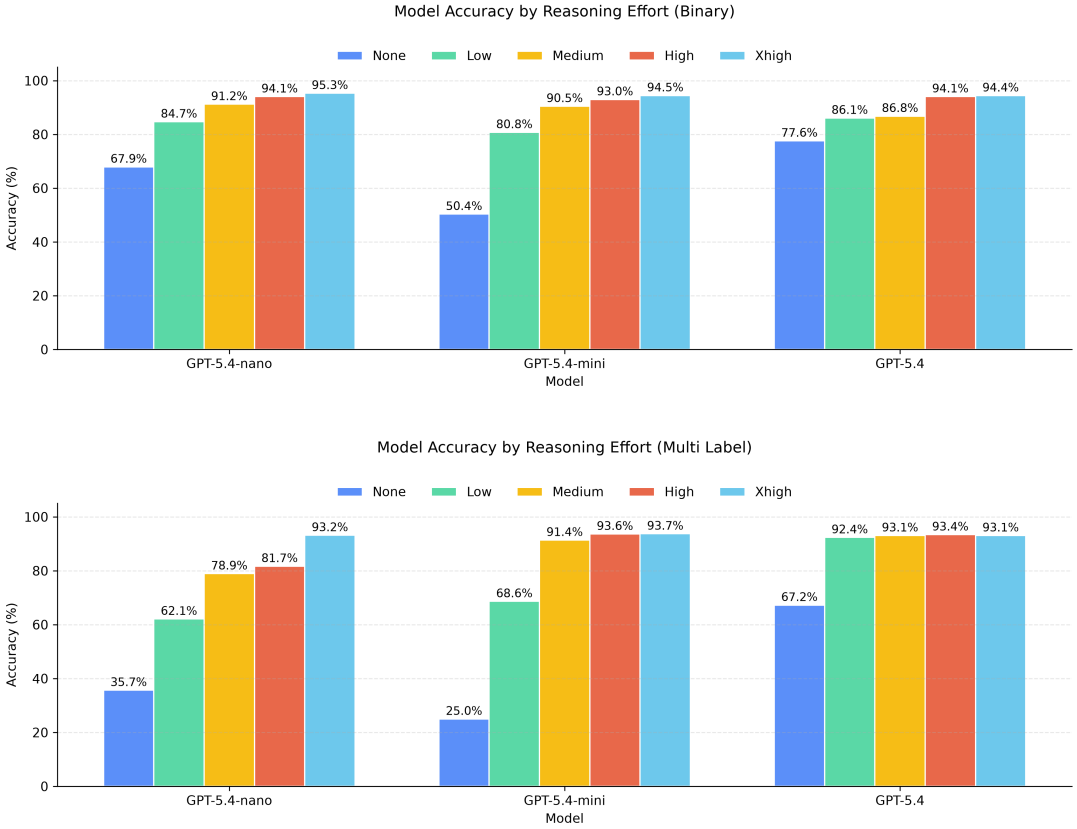


Figure 4: Mean LLM performances using different models and reasoning efforts on the binary task and multi-label task.

a difference between trials. We hypothesize that human participants may perform better over time, and that solving multi-label tasks may help them in solving the binary task. Based on the data we will collect in the pilot, we aim to investigate the optimal number of trials that could minimize these effects.

Another difference between humans and LLMs is their willingness to read large amounts of text. LLMs do not get tired or frustrated when passages of text are too long, but human participants might. The reasoning puzzles we intend to use in this study have a maximum of 10 relevant and 9 irrelevant arguments. Therefore, a single reasoning puzzle can contain up to 19 arguments. This might be too many for human participants to read and process. We will also evaluate this in the pilot study.

With this study, we aim to contribute to a better understanding of the reasoning capabilities and biases shared and not shared by humans and LLMs. Identifying different strengths and weaknesses can inform how tasks are best divided between humans and LLMs: if humans prove better at filtering out irrelevant arguments, for instance, it is worth investing human effort in filtering information before it reaches an LLM, whereas if LLMs handle this better, that filtering step is best delegated to the LLM. More broadly, these insights can help determine which parts of an argumentative reasoning task are best suited to human judgment or to LLM judgment, and possibly inform a combination of the two.

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